

Tribhuvan University
Institute of Science and Technology
2082



MDS / II Year / Third Semester/ Science
Data Science (MDS 601)
(Research Methodology)

Full Marks: 45
Pass Marks: 22.5
Time: 2 hours.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks. The symbols have their usual meanings.
Attempt all questions.

Group A

[5 × 3 = 15]

1. Define the term problem in research. Point out the criteria of good problem.
2. What do you mean by research design? Explain the essential elements of research design.
3. Point out the difference between "concept" and "construct".
4. What is ratio scale? What are its superiorities over other scales of measurements?
5. Explain the advantages and disadvantages of Systematic sampling.

Group B

[5 × 6 = 30]

6. Explain the characteristics of scientific method.
7. What do you mean by literature review? Illustrate the functions of literature review.
8. Mention different types of experimental research methods.
9. Define the term "validity". Describe the various types of validity.

OR

Define the term sampling. Elucidate the principal steps in sample survey.

10. Determine the minimum sample size required so that the sample estimate lies within 10% of the true value with 95% level of confidence when coefficient of variation is 60%.

OR

Explain the goal and structure of research proposal.

Tribhuvan University
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SCHOOL OF MATHEMATICAL SCIENCES

First Assessment 2081

Subject: Research Methodology

Course No: MDS 601

Level: MDS / II Year / III Semester

Full marks: 45

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Time: 2hrs

Candidates are required to give their answers in their own words as far as practicable.

Attempt All Questions.

Group A [5 × 3 = 15]

1. What are the different types of hypothesis? What is the main function of null hypothesis?
2. Explain the characteristics of scientific method.
3. Explain the difference between research method and Research methodology.
4. Define literature survey. What are the purposes of literature survey?
5. What are the steps involved in the research design process?

Group B [5 × 6 = 30]

6. What is scientific research? Describe the major steps involved in the scientific research process?
7. Define research problem. What steps are to be followed in identifying a research problem?
8. Explain the difference between a concept and a variable.
9. Explain the difference between fundamental research and applied research with suitable examples.

OR

What is case study research? Describe limitations and characteristics of a case study research.

10. Prepare a bibliographic entry in APA and IEEE method for the following :
- a) Book with one author
 - b) Journal article with two author

OR

A manager notices the number of grievances in the organization is increasing. The manager wishes to investigate this occurrence. What research design seems appropriate for the study? Give appropriate explanation.

Tribhuvan University
Institute of Science and Technology
SCHOOL OF MATHEMATICAL SCIENCES
Second Assessment 2082

Subject: Research Methodology
Course No: MDS 601
Level: MDS / II Year / III Semester

Full marks: 45
Pass Marks: 22.5
Time: 2hrs

Candidates are required to give their answers in their own words as far as practicable.

Attempt All Questions.

Group A [5x3=15]

1. Point out the difference between research methodology and research methods.
2. Explain the difference between reference and bibliography.
3. Explain nominal scale and interval scale with suitable examples.
4. Describe the APA method of citation for single author and multiple author with example.
5. Point out the characteristics and purpose of descriptive research design.

Group B [5x6=30]

6. What is the difference between a review of literature and a summary of literature?
7. Describe the major steps involved in the scientific research process?
8. Explain the concept and types of reliability.
9. What is research report? What purpose does it serve? Why is reporting of research study important?

OR

Explain the contents of research proposal. Is there any difference between a funded research proposal and an academic research proposal?

10. The mean systolic blood pressure of a certain group of people was found to be 125 mm of Hg with standard deviation of 15 mm of Hg. Calculate sample size to verify the result at 5 % level of significance if error do not exceed 2. Also find sample size if sample is selected from population of size 500.

OR

Distinguish between sampling and Non sampling error.

Tribhuvan University
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2079
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Master Level / Second Year / Third Semester / Science
Data Science (MDS 601)
(Research Methodology)

Full Marks: 45
Pass Marks: 22.5
Time: 2 hours

Candidates are required to give their answers in their own words as far as practicable.

Attempt All Questions.

Group A

[5 × 3 = 15]

1. What is applied research? How is it different from basic research?
2. What are the characteristics of scientific research? Is it possible to have good research that is not generalizable? Why or why not?
3. Why research problem is stated in a study? What are the criteria for a good problem statement?
4. What are the goals of a literature review? What distinguish a strong from a weak literature review?
5. Describe the nature of descriptive research. What is the value of descriptive research in data science?

Group B

[5×6=30]

6. What is scientific research? Describe the major steps involved in the scientific research process?
7. What type of experiment do you think it would be possible to conduct in the following situation? Describe the different steps that you would take in conducting the experiment. "A company has designed a new product. You think that the consumer will like this product more than other similar products. You wish to demonstrate by scientific experiment, that this is the case."
8. "A valid instrument is always reliable, but a reliable instrument may not always be valid" comment on this statement.
9. What is non-response error? Describe how non response error can occur? What can be done to rescue the non-response errors associated with mail questionnaire?

OR

Describe sampling. Why is sampling used in research? List the steps involved in the process of sampling.

10. What factors should be kept in mind while selecting a research topic? Explain the attributes of a good research topic.

OR

What is a research report? What purposes does it serve? Why is reporting of research study important?

Tribhuvan University
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2081
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Master Level / Second Year /Third Semester/ Science
Data Science (MDS 601)
(Research Methodology)

Full Marks: 45
Pass Marks: 22.5
Time: 2 hours

Candidates are required to give their answers in their own words as far as practicable.

Attempt All Questions.

Group A

[5 × 3 =15]

1. Define research. Describe the main characteristics of scientific research.
2. Describe the nature and purpose of historical research.
3. Describe the APA method of citation for single author and multiple author with example.
4. Illustrate Nominal scale and Ratio Scale.
5. Why research problem is started in a study? What are the criteria for a good research problem?

Group B

[5×6=30]

6. Define the term "Validity". Describe the various types of validity.
7. Illustrate the major steps involved in scientific research process.
8. Mention the principal steps in a sample survey.
9. Under what condition a laboratory experiment be better than field experiment.

OR

What is sampling error? How sampling error can be reduced during the process of data collection.

10. What library skills are required for thorough survey of literature related to a research topic in Data science.

OR

In Measuring reactions time, a psychologist estimates that the standard deviation is 0.05 seconds. How large a sample of measurement must be taken in order to be 99% confident that the error of his estimate will not exceed 0.01 seconds?

Tribhuvan University
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SCHOOL OF MATHEMATICAL SCIENCES
Second Assessment 2081

Subject: Research Methodology
Course No: MDS 601
Level: MDS / II Year / III Semester

Full marks: 45
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Time: 2hrs

Candidates are required to give their answers in their own words as far as practicable.
Attempt All Questions.

Group A [5x3=15]

1. What is bias how can it be reduced while interviewing with respondent?
2. Explain how stratified sample can reduce the standard error?
3. Describe the nature and purpose of case study design?
4. Define literature review. What are the benefits of good literature survey?
5. Explain the ordinal and ratio scale with suitable examples.

Group B [5x6=30]

6. A research scholar has to work as a judge and derive the truth not as a Pleader who is only eager to prove his/her case in favor of his/her plaintiff. Discuss the statement pointing out the objective of research.
7. A manager notices the number of grievances in the organization is increasing. The manager wishes to investigate this occurrence. What research design seems appropriate for the study? Give appropriate explanation.
8. Under what condition might a field experiment be better than a laboratory experiment?
9. Define the term reliability. Discuss the various types of reliability.

OR

Discuss the relative advantages and disadvantages of questionnaire and schedule.

10. What are the various methods of documenting sources? Describe the APA method of citation with examples.

OR

What is a research proposal? Point out the different steps in research proposal with brief explanation.

Tribhuvan University
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SCHOOL OF MATHEMATICAL SCIENCES
First Assessment 2080

Subject: Research Methodology
Course No: MDS 601
Level: MDS / II Year / III Semester

Full marks: 45
Pass Marks: 22.5
Time: 2hrs

Candidates are required to give their answers in their own words as far as practicable.
Attempt All Questions.

Group A [5x3=15]

1. Explain the difference between fundamental research and applied research with suitable examples.
2. What is scientific research? Describe the major steps involved in the scientific research process?
3. Define literature survey. What are the benefits of a good literature survey?
4. What are the steps involved in the research design process?
5. What is ratio scale? What are its superiorities over other scale of measurement

Group B [5x6=30]

6. Define the term reliability. Describe the various types of reliability.
7. What is case study research? Describe the steps and characteristics of a case study research.
8. Describe the difference between statistical hypothesis and Research hypothesis.
9. Describe sampling. Why is sampling used in research? List the steps involved in the process of sampling.

OR

What is non-sampling error? How non-sampling error can be reduced in the process of data collection?

10. prepare a bibliographic entry in APA method and MLA method for the following :
- a) for a book with one author
 - b) for a journal article

OR

Describe experimental research method. Point out the mandatory Steps in Experimental Methods.

Tribhuvan University
Institute of Science and Technology
SCHOOL OF MATHEMATICAL SCIENCES
Second assessment 2079

Subject: Research Methodology
Course No: MDS 601
Level: MDS/II Year/III Semester

Full Marks: 45
Pass Marks: 22.5
Time: 2hrs

Candidates are required to give their answer in their own words as far as practicable.

Attempt ALL questions.

Group A [5×3=15]

1. What is case study? Describe the characteristics of case study.
2. Describe the nature and purpose of historical research.
3. What is ratio scale? What are its superiorities over other scale of measurement
4. Explain how a stratified sample can reduce the standard error?
5. What is research report? What purposes does it serve?

Group B [5x6=30]

6. A manager notices the number of grievances in the organization is increasing. The manager wishes to investigate this occurrence. What research design seems appropriate for the study? Give appropriate explanation.
7. Under what conditions might be a laboratory experiment be better than a field experiment.
8. Define the term "reliability" and "validity". Describe the various types of validity.
9. What is a questionnaire? What are the most critical problems or concerns in questionnaire design?

OR

What are the sampling error and non-sampling error? How non-sampling error can be reduced in the process of data collection?

10. Explain the conventions of academic writing.

OR

What are the various methods of documenting sources? Describe the **APA** method of citation with examples

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Tribhuvan University
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SCHOOL OF MATHEMATICAL SCIENCES
First Assessment 2079

Subject: Research Methodology

Course No: MDS 601

Level: MDS /II Year /III Semester

Full Marks: 45

Pass Marks: 22.5

Time: 2hrs

Candidates are required to give their answers in their own words as far as practicable.

Attempt All Questions.

Group A [3 × 5 = 15]

1. Explain the difference between research methods and Research methodology.
2. What do you mean by Research Hypothesis? Explain in brief with one case study.
3. Define Inductive and Deductive theory with suitable examples.
4. Distinguish between research hypothesis and statistical hypothesis.
5. Define literature review. What are the benefits of good literature survey?

Group B [5 × 6 = 30]

6. "The distinction between basic and applied research is that the purpose of the former is to answer a question and the purpose of the later is to solve a question". Comment on this statement.
7. What is scientific research? Describe the major steps involved in the scientific research process?
8. What library skills are required for a thorough survey of literature related to a research topic in Data science?
9. Discuss research design. Is it possible to conduct a research study without a formal research design? How?

OR

Define descriptive research design? What are the characteristics of descriptive research design?

10. Describe a research design for testing the following hypothesis. "Male managers will have more access to critical information than female managers in the same ranks".

OR

What are the elements of research design? Point out the initial questions in developing a research design.

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Time: 2hrs

Candidates are required to give their answers in their own words as far as practicable.
Attempt All Questions.

Group A [5x3=15]

1. ✓ What is scientific research? Describe the major steps involved in the scientific research process?
2. ✓ What are the steps involved in the research design process?
3. ✓ Describe the difference between statistical hypothesis and Research hypothesis.
4. ✓ Explain the difference between fundamental research and applied research with suitable examples.
5. ✓ Define literature survey. What are the benefits of a good literature survey?

Group B [5x6=30]

6. ✓ What is case study research? Describe the steps and characteristics of a case study research.
7. ✓ Define historical research. Describe the nature and characteristics of historical research.
8. ✓ Why research problem is stated in a study? What are the criteria for a good problem statement?
9. ✓ Many researchers feel that qualitative research is mainly for generating the hypothesis rather than for testing it. Give your critical comments on this.

OR

Explain the meaning of "concept" and "construct". How can you differentiate between the two?

10. ✓ Prepare a bibliographic entry in APA 7th version method for the following :
- a) for a book with one author
 - b) for a journal article

OR

Describe experimental research method. Point out the mandatory Steps in Experimental Methods

Tribhuvan University
Institute of Science and Technology
SCHOOL OF MATHEMATICAL SCIENCES
First Re-Assessment-2082

Subject: Research Methodology
Course No: MDS 601
Level: MDS / II Year / III Semester

Full marks: 45
Pass Marks: 22.5
Time: 2hrs

Candidates are required to give their answers in their own words as far as practicable.
Attempt All Questions.

Group A [5x3=15]

1. ✓ Define the term "hypothesis". Mention the function of hypothesis in brief.
2. ✓ What are the steps involved in the research design process?
3. ✓ Explain the difference between fundamental research and applied research with suitable examples.
4. ✓ Define literature survey. What are the benefits of a good literature survey?
5. ✓ Define the term problem in research. Point out the criteria of good problem.

Group B [5x6=30]

6. ✓ What is scientific research? Describe the major steps involved in the scientific research process?
7. ✓ Describe the difference between statistical hypothesis and Research hypothesis.
8. ✓ What is case study research? Describe the steps and characteristics of a case study research.
9. ✓ Define historical research. Describe the nature and characteristics of historical research.

OR

Explain the meaning of "concept" and "construct". How can you differentiate between the two?

10. ✓ Prepare a bibliographic entry in APA 7th version method for the following :
- a) for a book with one author
 - b) for a journal article

OR

Describe experimental research method. Point out the mandatory Steps in Experimental Methods
